

Mesith Rathnayake

B.Sc. (Eng) in Computer Engineering (UG)

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📍 No 97/A/1/2, Wataraka East, Galle 🌐 Mesith Rathnayake 🔄 Mesit Rathnayake

SUMMARY

Computer Engineering undergraduate at the University of Ruhuna with hands-on experience in software development and exploratory projects in machine learning and computer vision. Familiar with image processing techniques, data handling, and building simple ML-based applications. Passionate about developing intelligent and efficient solutions. Seeking a Machine Learning or Computer Vision & Image Processing internship to apply and further strengthen practical skills in a real-world environment.

PROJECTS

Book Clustering using Machine Learning (Hands-on Learning Project) 🔗

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn

- Designed and implemented an unsupervised learning pipeline to cluster books based on popularity and engagement metrics, gaining hands-on experience with real-world data analysis.
- Performed data preprocessing including handling missing values, feature engineering (publication year extraction), and feature scaling using StandardScaler to ensure consistent model performance.
- Applied clustering algorithms such as K-Means and Agglomerative Clustering to group similar books and analyze underlying data patterns.
- Determined the optimal number of clusters using Silhouette Score and evaluated model performance using Davies-Bouldin Index and Within-Cluster Sum of Squares (WCSS).
- Visualized high-dimensional data using PCA to interpret cluster distributions and support comparative analysis between clustering approaches.

Classroom Engagement Monitoring System 🔗

Tech Stack: Python, Streamlit, OpenCV, dlib, NumPy, SciPy, computer vision, facial landmark detection.

- Built an exploratory computer vision system to analyze student engagement using facial landmark detection techniques.
- Implemented basic attention estimation using Eye Aspect Ratio and head pose concepts with OpenCV and dlib.
- Developed a simple Streamlit interface to visualize live video feed and engagement-related metrics.

BudgetBuddy – DevOps Implementation (Hands-on Learning Project) 🔗

Tech Stack: React, Node.js, Express.js, MongoDB, Docker, Docker Compose, Jenkins, Terraform, Ansible, AWS EC2, GitHub, Docker Hub, Grafana, Prometheus

- Designed and deployed a full-stack application to gain hands-on experience with DevOps practices, including CI/CD pipelines, Infrastructure as Code, and production monitoring.
- Implemented a Jenkins CI/CD pipeline triggered by GitHub webhooks to automate build, containerization, and deployment processes to AWS EC2.
- Integrated monitoring and health-check workflows to track application availability and improve incident visibility during deployments.
- Reduced manual deployment effort by automating build and release workflows, improving consistency, reliability, and recovery speed.
- Utilized Docker and Docker Compose for containerization, Terraform for infrastructure provisioning, and monitoring tools to support repeatable and observable deployments.

MOH Connect - Maternal & Child Care Management System (Group Project - In Progress)

Tech Stack: React.js, JavaScript, React Router, Tailwind CSS, HTML, CSS

- Contributing as a frontend developer in a group project focused on building a role-based healthcare management system.
- Developing UI components and collaborating with the team to integrate backend APIs for authentication and healthcare workflows.

LEADERSHIP & VOLUNTEERING EXPERIENCE

District Director - Member Relations & Fellowship

07/2025 – Present

Leo District 306 D8 | Lions International

- Oversee member engagement, fellowship activities, and inter-club coordination across the district
- Contribute to strategic planning and leadership development initiatives

Junior Treasurer

02/2026 – Present

REF Media - Faculty of Engineering, University of Ruhuna

- Support financial planning and budgeting for media-related events and productions
- Coordinate videography and live-streaming operations for faculty-level events
- Assist in managing media teams, production workflows, and technical setups for live broadcasts

Design Team Lead

- ReXtro 2025 - Silver Jubilee Exhibition, Faculty of Engineering, University of Ruhuna
- Red Cypher 2.0 - Inter-university Capture the Flag Competition
- IEEEExtreme 18.0 - Global Programming Competition (Representing IEEE SB UOR)
- Led creative direction, branding, and visual design for large-scale technical events
- Coordinated design teams to deliver consistent and impactful visual identity

EDUCATION

B.Sc. (Engineering) [Hons], Computer Engineering

05/2023 – Present

Faculty of Engineering, University of Ruhuna

G.C.E Advanced Level

01/2018 – 08/2021

Richmond College

Physical Science - AAA

G.C.E Ordinary Level

02/2007 – 12/2017

Richmond College

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SKILLS

- DevOps & Cloud Tools (Docker, Jenkins, Terraform, AWS)
- Full-Stack Web Development (React, Node.js, MongoDB)
- Mobile Application Development (Kotlin, Firebase)
- Creative Design, Media Production & Team Coordination

ACHIEVEMENTS

Secured a Top 8 position at Eminence 5.0

Ranking among the Top 10 teams overall, competing in Power System Simulation (MATLAB), Network Design (Packet Tracer), Machine Learning, and Innovation Project Pitching as part of Team Quadra.

Second Runners-Up at VoltCast 1.0 Competition

IEEE PES Student Branch and the Electrical & Information Engineering Society, University of Ruhuna (Team ScorpionX).

REFERENCES

Dr. Prabath Weerasinghe, Senior Lecturer,

Department of Electrical & Information Engineering, FoE UoR

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